7. D7 — One-page Engineering Recommendation

Case: T6 — Micro-lending / SHG Credit Management System

The selected system manages SHG/member information, loan applications and approvals, repayment tracking, financial balances, reporting, authorization, and audit history. The key characteristics are high financial risk, strong security and authorization needs, auditability, and very low tolerance for incorrect balances or repayment status. Requirements may change as stakeholders refine workflows, but correctness remains the dominant engineering concern.

Decision method. The nine common criteria were weighted to 100%. Risk exposure (22%) and defect tolerance (20%) received the highest weights because incorrect loan, balance, repayment, or authorization data can directly affect financial decisions. Regulatory burden received 16% because controlled access and traceability are essential. The remaining weight covers requirements volatility, project size, team distribution, customer availability, contract type, and delivery urgency.

Recommendation: V-Model. The V-Model achieved 4.22/5, ahead of Scrum at 4.20/5 and Spiral at 4.18/5. Its main advantage is disciplined verification and validation throughout development. Requirements can be traced to design and tests, while system-level acceptance can explicitly validate calculations, approvals, balances, access controls, and audit trails.

Why not Spiral? Spiral is an excellent runner-up because it emphasizes risk management and iterative refinement. However, this is a semester-sized academic project and the dominant challenge is dependable correctness and control rather than high technology uncertainty. V-Model therefore gives stronger rigor with manageable process overhead.

Why not Scrum? Scrum is strong when requirements change frequently and stakeholder feedback must drive short iterations. In the baseline case, requirements volatility is medium while risk, regulatory burden, and defect tolerance dominate. Sensitivity tests show Scrum becomes the winner when any one of those three high-priority weights is reduced by 5 points and the weight is transferred to requirements volatility, so the recommendation is conditional.

Engineering approach. Use V-Model planning with requirement baselines, traceability from requirements to tests, unit/integration/system/acceptance testing, and documented approval criteria. Apply role-based authorization, strong authentication, least privilege, protected financial data, and audit logging. Every financial calculation and state transition should have explicit test cases; critical defects should block acceptance until resolved.

Final recommendation: V-Model is the best-fit SDLC model for T6 under the stated weights. The selection is driven by dependable financial correctness and verification, not by assuming Scrum is always superior. Revisit the model if requirements become highly volatile or the project shifts toward frequent incremental releases; under those conditions, Scrum becomes more appropriate.